

in real time. It can then be easily made available to the water board authority, engineers and consumers through bills, or on the Internet via Web portals and apps. Analytics aids in extracting useful information and improving efficiency.

Along with the residential sector, smart meter deployment is rising in commercial and industrial sectors also. Key players such as Itron, Xylem and

Maven Systems are expanding their presence by entering new markets.

For further adoption of smart meters, several challenges need to be considered. No other person knows the schedule of the consumer as meters do not check which appliance is being used at a particular time. Also, while taking readings from a remote location, radiations of low energy levels are emitted and are, hence, less harmful.

Despite this, security and radio-waves emissions are seen as potential risks when it comes to smart meters.

Initial investment and operating costs are high. Proper technical standards and regulations should be established to ensure smooth deployment. Once existing meters become obsolete, jobs must be redesigned to cater to the new necessities.

—Ayushee Sharma

3 RFID PCB TAGS CREATING AND PROTECTING VALUE

RFID tags solve the challenges of barcodes by simultaneously reducing manual operations in the process

Radio frequency identification (RFID) tags provide reliability, higher reading accuracy, wide read angle, flexibility and read-through package. Information stored can be altered and occupies less space on the PCB (by replacing multiple barcode labels) due to its simplicity and fewer manual operations than barcodes.

RFID is rewritable, and can also work without an optical connection through carton box, product housing, etc. It covers all uses over the full value chain. Shorter wavelength of UHF results in minimal antenna space needed on PCBs. A device like Magicstrap can be added to design smart PCBs with RFID tag function. It enables products to be fully-traceable within the supply chain.

RFID is a battery-less RF technology with ultra-high frequency (UHF) range of 860MHz to 960MHz, which can cover not only a distance greater than five metres but also short-range communication. It has several advantages over barcode, such as cost saving, faster data capture (very short read time) and reading multiple tags at once (more than hundred tags per second). RFID has overcome various challenges faced by businesses like piracy, traceability and wrong delivery of products, in addition to multiple manual operations.

An RFID system has three elements, namely, tags, antenna and software system, and works in seven

steps, as follows:

- Tag enters RF field of reader
- RF signal powers tag
- Tag transmits ID plus data
- Reader captures data
- Reader sends data to computer
- Computer sends data to reader
- Reader transmits data to tag

Shafi Ahmed, product engineer, Murata Electronics India Pvt Ltd, says “Magicstrap solves the challenges of tracking and after-sales service in a mass-manufactured electronic products scenario.

“By using a standard RFID reader, users can trace the products from factory or warehouse to final shipments. They can design the software and GUI based on their needs and requirements. The reader has numerous other applications in asset tracking, anti-counterfeiting, retail and more.”

Magicstrap contains an RFID IC, an antenna filter to cover worldwide RFID frequencies, an antenna matching circuit made of low-temperature cofired ceramics and electrostatic discharge protection (< 2kV/mm).

The antenna only receives incoming RF; all other functions like impedance matching, fixed-centre frequency and bandwidth are provided by Magicstrap.

There are four types of reference antennae on the PCB. Change of board size only affects communication range and not functionality.

Materials for making PCB tags are eco-friendly and hundred per cent green. Tags connect all elements of

the chain—R&D, production, logistics, distribution, point-of-sales and marketing, aftermarket and recycling. These provide near-field transmission with loop antenna, for which the user only needs to control the process.

There are four recommended antenna patterns to a PCB when using a device like Magicstrap. To gather practical experience or test possible performance before touching the layout of the PCB and adding one of these patterns permanently, a mini PCB can be used.

A mini PCB should ideally be used at the location where the user plans to later integrate Magicstrap antenna pattern in the layout. Magicstrap should be mounted as close as possible to the middle of the PCB so that maximum read range can be achieved. The same is valid for the mounting position of the mini PCB.

To communicate with only one Magicstrap, it is important to make sure that there is no other RFID tag present in the field of the reader. To test bulk-read function, the user can place as many tags in the field as needed.

RFID is compatible with EPC Global standard C1G2 and with any EPC Global C1G2-compliant equipment. It carries 240-bit EPC for serial and additional 512-bit memory. It enables users to freely add their own information (production date, version ID, etc), while keeping their own numbering system internally. It is ISO 18000-6-compliant and UHF air-interface-standardised.

—Deepshikha Shukla